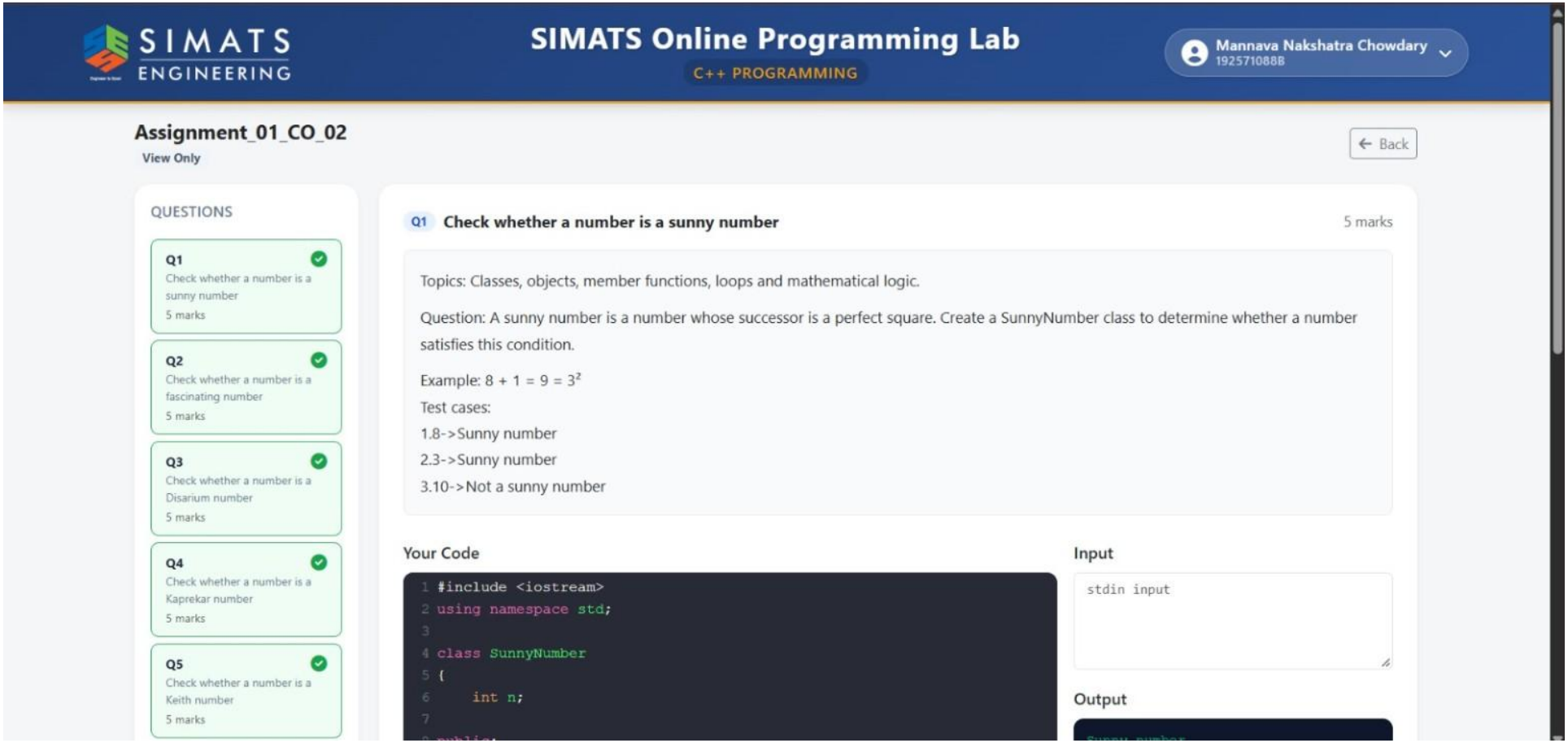
DSA0118

ASSIGNMENT 01 – CO2

Q1 – Sunny Number: Question



Q1 – Sunny Number: Code and Output

SIMATS ENGINEERING

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C++ PROGRAMMING

Mannava Nakshatra Chowdary
192571088B

Q4
Check whether a number is a Kaprekar number
5 marks

Q5
Check whether a number is a Keith number
5 marks

Q6
Check whether a number is a Trimorphic number
5 marks

6/6 answered

Your Code

```cpp
#include <iostream>
using namespace std;

class SunnyNumber
{
    int n;

public:
    void getData()
    {
        cin >> n;
    }

    void check()
    {
        int num = n + 1;
        int i = 1;

        while (i * i < num)
        {
            i++;
        }

        if (i * i == num)
            cout << "Sunny number";
        else
            cout << "Not a sunny number";
```

Input

stdin input

Output

```
Sunny number
```

Visible Test Cases

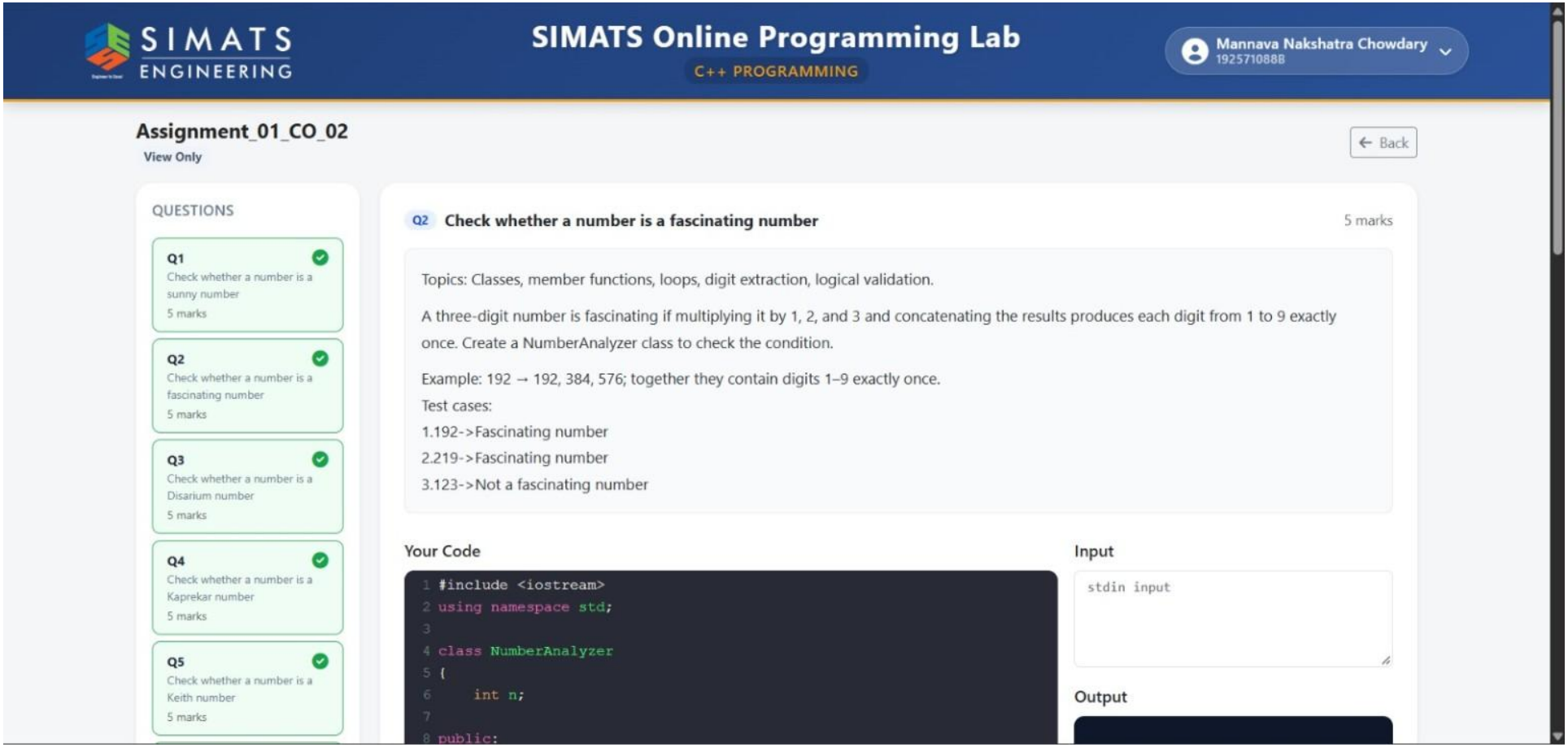
Test Case 1 ✓

Test Case 2 ✓

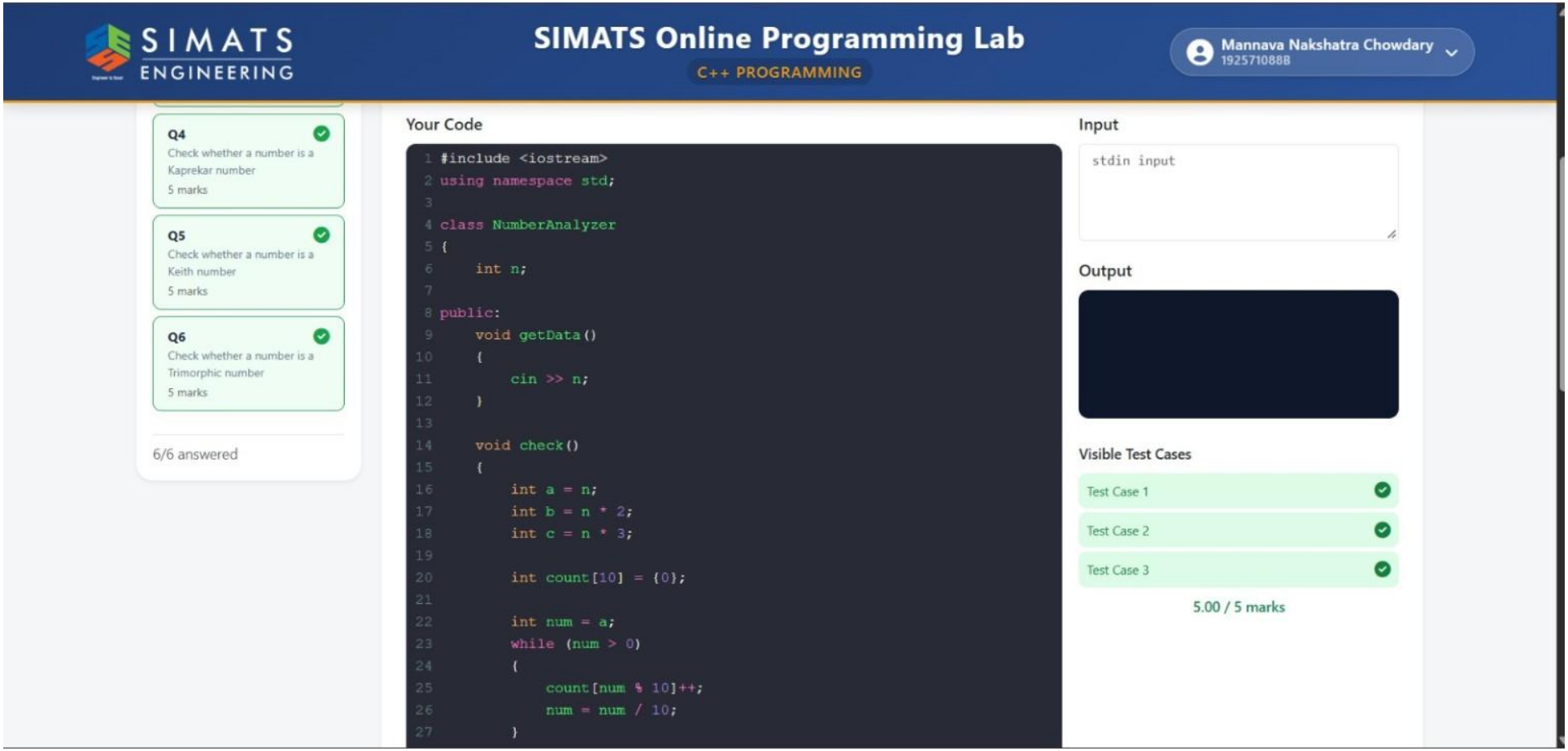
Test Case 3 ✓

5.00 / 5 marks

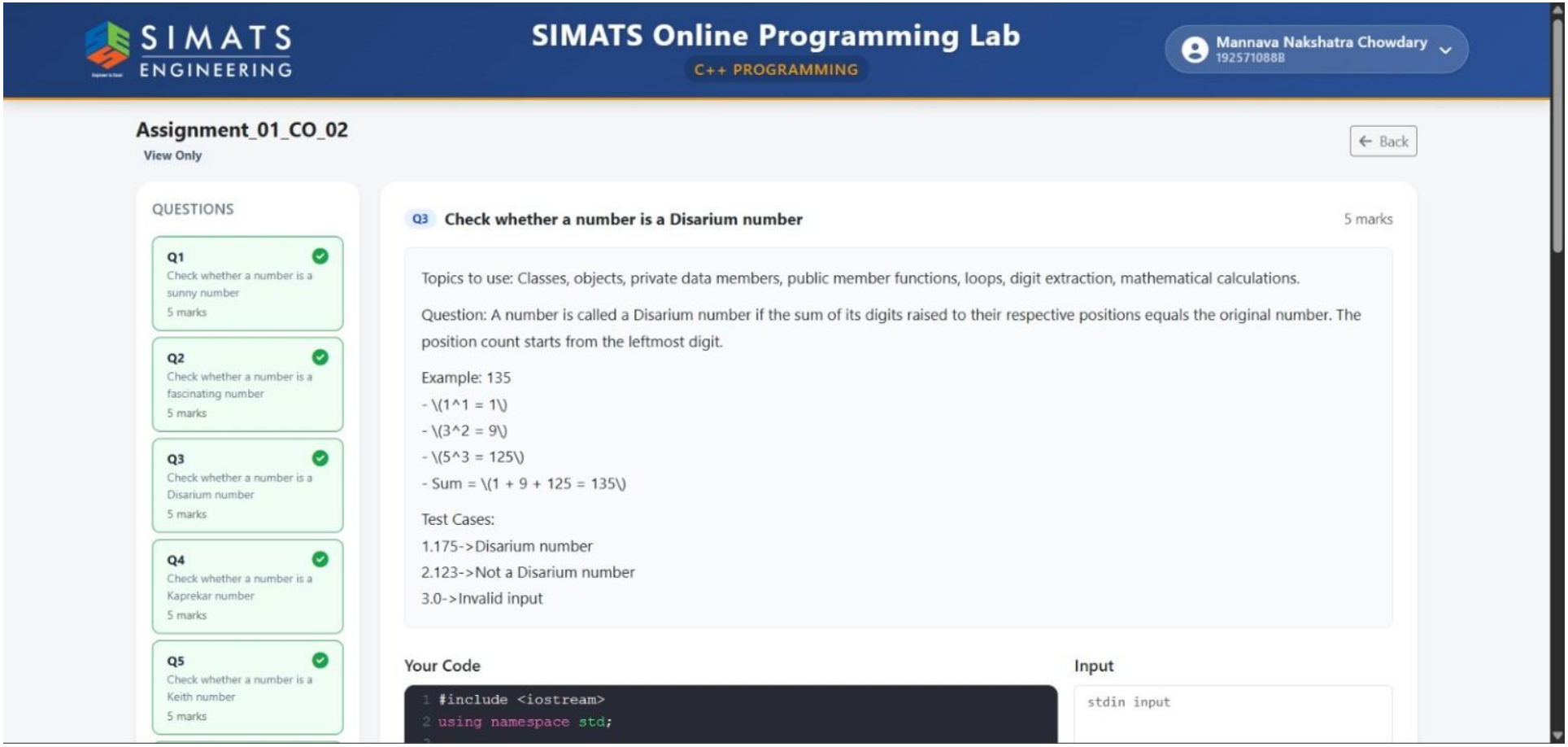
Q2 – Fascinating Number: Question



Q2 – Fascinating Number: Code and Test Cases



Q3 – Disarium Number: Question



Q3 – Disarium Number: Code and Output

SIMATS ENGINEERING

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C++ PROGRAMMING

Mannava Nakshatra Chowdary
1925710888

Q5
Check whether a number is a Keith number
5 marks

Q6
Check whether a number is a Trimorphic number
5 marks

6/6 answered

Your Code

```cpp
#include <iostream>
using namespace std;

class Disarium
{
    int n;

public:
    void getData()
    {
        cin >> n;
    }

    void check()
    {
        if (n <= 0)
        {
            cout << "Invalid input";
            return;
        }

        int temp = n;
        int digits = 0;

        while (temp > 0)
        {
            digits++;
```

Input

stdin input

Output

```
Invalid input
```

Visible Test Cases

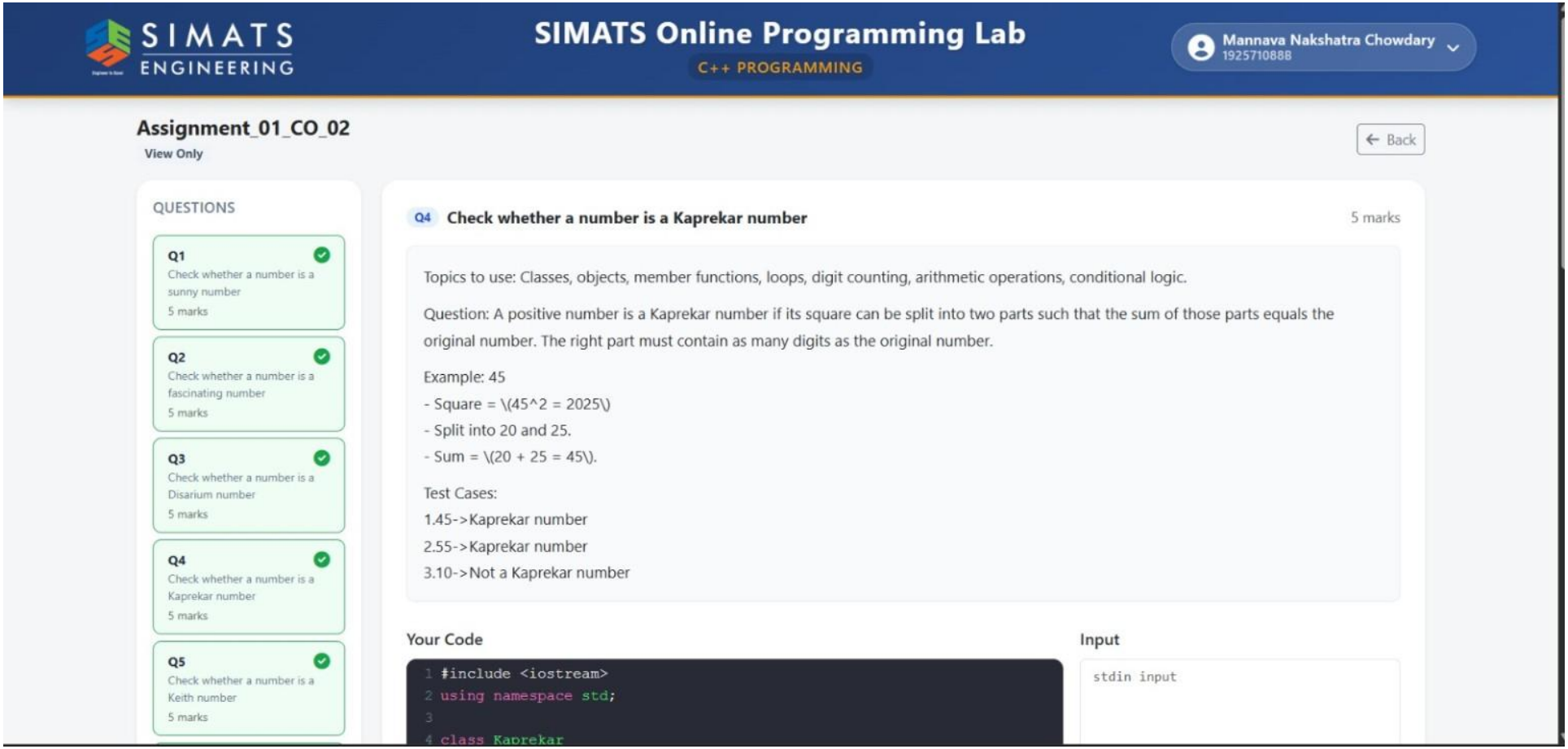
Test Case 1 ✓

Test Case 2 ✓

Test Case 3 ✓

5.00 / 5 marks

Q4 – Kaprekar Number: Question



Q4 – Kaprekar Number: Code and Output

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C++ PROGRAMMING

Mannava Nakshatra Chowdary
1925710888

Q5
Check whether a number is a Keith number
5 marks

Q6
Check whether a number is a Trimorphic number
5 marks

6/6 answered

Your Code

```cpp
#include <iostream>
using namespace std;

class Kaprekar
{
    int n;

public:
    void getData()
    {
        cin >> n;
    }

    void check()
    {
        if (n <= 0)
        {
            cout << "Invalid input";
            return;
        }

        long long square = (long long)n * n;

        int temp = n;
        int digits = 0;

        while (temp > 0)
        {
```

Input

stdin input

Output

Invalid input

Visible Test Cases

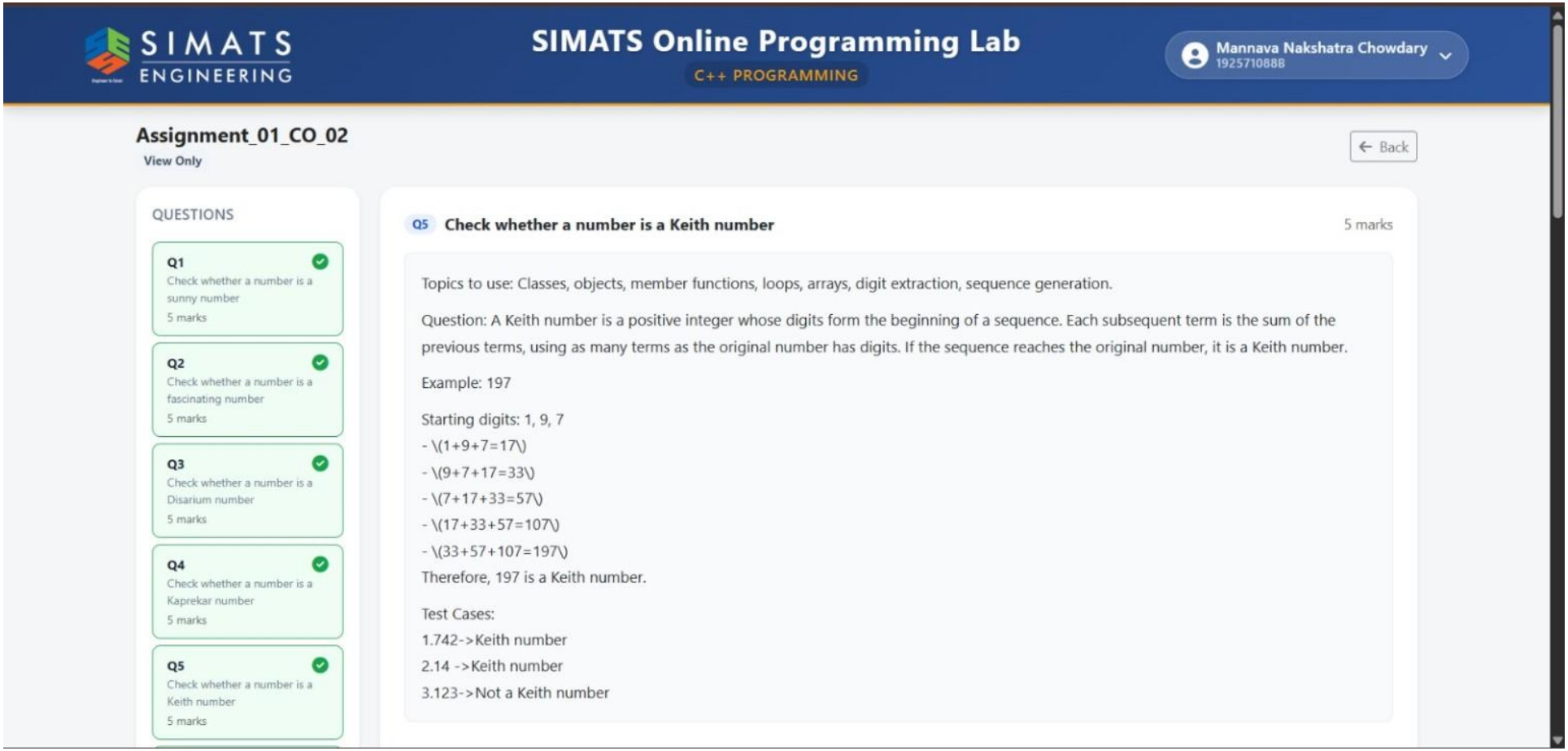
Test Case 1

Test Case 2

Test Case 3

5.00 / 5 marks

Q5 – Keith Number: Question



Q5 – Keith Number: Code and Output

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C++ PROGRAMMING

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Q6
Check whether a number is a Trimorphic number
5 marks

6/6 answered

Your Code

```
#include <iostream>
using namespace std;

class KeithNumber
{
    int n;

public:
    void getData()
    {
        cin >> n;
    }

    void check()
    {
        if (n <= 0)
        {
            cout << "Invalid input";
            return;
        }

        int temp = n;
        int digits[20];
        int count = 0;

        while (temp > 0)
        {
```

Input

stdin input

Output

Invalid input

Visible Test Cases

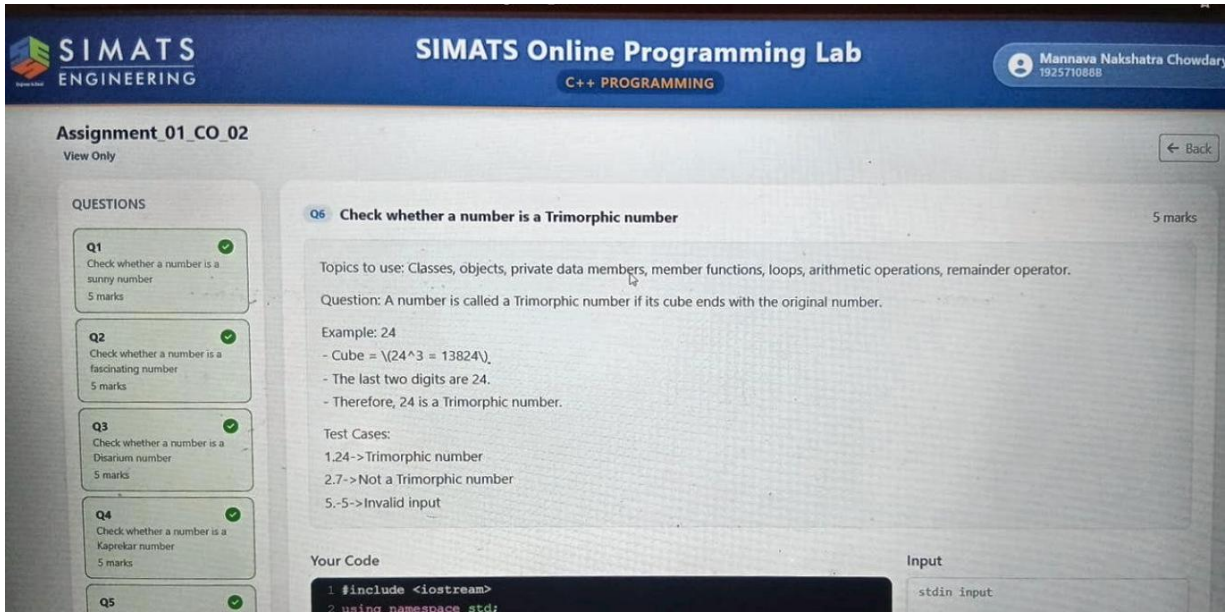
Test Case 1

Test Case 2

Test Case 3

5.00 / 5 marks

Q6 – Trimorphic Number



Q6 – Trimorphic Number: Code and Output

